Check for updates

Tackle Techniques and Characteristics Associated With a Concussion in Tackling Players in the National Football League

Christopher P. Sherwood,*† MS, Fintan Grogan,‡ BSc, Timothy L. McMurry,§ PhD, James R. Funk,† PhD, PE, Jeff R. Crandall,† PhD, Allen Sills,‖ MD, and Gregory Tierney,¶ PhD
Investigation performed at Biomechanics Consulting and Research, Charlottesville, Virginia, USA

Background: Concussions remain a major concern in football. A qualitative video review can provide contextual evidence supporting changes in rules and coaching techniques aimed at reducing the risk of players sustaining a concussion.

Purpose: To identify tackling techniques and characteristics associated with concussions to the tackling player.

Study Design: Case-control study; Level of evidence, 3.

Methods: A qualitative review of 51 concussions sustained by tacklers in National Football League (NFL) games between the 2015 and 2019 seasons as well as 96 control tackles was performed using video from multiple perspectives. For each concussive case, 1 to 3 nonconcussive control tackles were matched based on player trajectory and closing speed as determined from on-field player tracking data from Next Gen Stats. Cases and controls were coded for 20 different tackle techniques and characteristics. The data were analyzed using conditional logistic regression models, and the results were expressed as odds ratios (ORs) with 95% confidence intervals (CIs) as well as adjusted and unadjusted *P* values.

Results: Despite attempting to match cases and controls based on closing velocity, closing velocity was significantly associated with the concussion risk in every logistic regression model. Predictors that had the greatest effect on the increased risk of concussions were ''tackler leading with helmet'' (OR, 5.18 [95% CI, 1.87-14.33]; false discovery rate [FDR]–adjusted $P = .0032$) and ''tackler primary loading'' being the helmet as opposed to the shoulder (OR, 4.61 [95% CI, 1.41-15.05]; FDR-adjusted $P = .0000$). Important factors associated with a reduced concussion risk were the tackler's ''head placed on correct side'' of the ball carrier (OR, 4.17 [95% CI, 1.58-11.01]; FDR-adjusted $P = .0088$) and ''ball carrier primary loading,'' with the torso having the lowest risk relative to all other body segments.

Conclusion: The less a tackler involved his helmet in a tackle, the lower his risk of sustaining a concussion. This study supports teaching players to place their heads on the correct side of the ball carrier (neither into the ball carrier nor into the path of the ball carrier) when tackling to reduce their risk of a concussion, and players should target the torso of the ball carrier as their primary contact point.

Keywords: football; video review; concussion; injury prevention

A concussion is a major health concern in many sports, particularly high-contact sports such as rugby and football. Efforts to reduce sports concussions have focused on 2 complementary approaches: changes to equipment and changes to player behavior. Changes to equipment, such as helmets and padding, are passive safety measures that reduce the severity of head impacts when they occur. Changes to player behavior through player education, coaching techniques, and rule changes constitute active safety measures that seek to prevent severe head impacts from occurring in the first place. Since 2002, the National Football League (NFL) has implemented more than 50 rule changes to increase safety within the game.[15] Because of rule changes and other factors, the source of concussions in NFL players has altered dramatically over the years. Helmet-to-helmet collisions accounted for 68% of concussions in the 1996 to 2001 seasons[17] but only 36% of concussions in the 2015 to 2017 seasons.[12] Currently, nearly two-thirds of concussions in the NFL are sustained during tackling, with the tacklers sustaining nearly twice as many concussions as the ball carriers.[12]

NFL games receive extensive video coverage from multiple high-quality cameras positioned around the stadium, making a video review a good choice for examining concussions in the NFL. A video review of sports matches has been used to analyze the causes of concussions in ice hockey,[10] soccer,[7] rugby union,[9,20] and the NFL.[12,17] In

The American Journal of Sports Medicine
1–10
DOI: 10.1177/03635465251321005
© 2025 The Author(s)

these studies and the current study, the video review was qualitative, meaning that reviewers made categorical determinations rather than videogrammetric measurements. Qualitative video reviews have been used extensively to examine tackling techniques in rugby union, which utilizes tackling techniques similar to football.[2,9,20,21,25] These studies have identified 4 patterns of tackler behavior associated with a reduced rate of head injuries for the tackler: (1) approaching with the "head up and forward/face up"[22]; (2) shortening steps before contact[22]; (3) bending at the waist[25]; and (4) placing the head on the "correct" side of the ball carrier, away from the ball carrier's forward momentum.[21,22] Qualitative video reviews of football games have shown that tackling with the head up and the helmet's facemask on the correct side of the ball carrier (toward the inside shoulder) results in more successful tackles than tackling with the head down or "across the bow."[13,19] Even though it is considered more dangerous in rugby and has been shown to be less successful in football, many coaches currently teach the "head across the bow" tackling technique.[19]

The goal of this study was to identify tackling techniques and characteristics associated with concussions to the tackling player. These findings may translate into new coaching strategies, educational opportunities, and rule changes that reduce the incidence of concussions to tacklers in the future.

METHODS

Study Design

Concussions sustained by NFL players during games are diagnosed by club physicians in accordance with the NFL Head, Neck and Spine Committee's concussion diagnosis and management protocol, which is collectively bargained and agreed upon by the NFL and the NFL Players Association.[11] All concussions sustained by NFL players during games and practices are recorded in the NFL's league-wide electronic medical record system. The NFL electronic medical record system was collectively bargained between the NFL and the NFL Players Association to serve as a workplace injury recording system.[14] NFL players sign authorization forms for the data provided to this system to be used in an aggregated, deidentified fashion in furtherance of certain research approved by the league and the union. This study underwent the approval process and was approved by the NFL and NFL Players Association.

A qualitative video review was conducted in this matched case-control study to compare tackling techniques and characteristics between concussed (case) and nonconcussed (control) tacklers engaged in similar tackles in terms of closing speed and player trajectory. There were 20 different tackling characteristics compared (Table 1), and a false discovery rate (FDR) threshold of 0.10 was used to control for multiple comparisons.

Data Collection

A tackle was defined by the NFL Rulebook (rule 3-35) as "an attempt by a defensive player to hold a runner to halt his advance or bring him to the ground."[16] The video review in this study was limited to pass and rush plays. Special teams plays and concussions occurring after a turnover were excluded because these plays can have more of a chaotic nature. All defensive positions were eligible as the tackling player, whereas all offensive positions with the exception of offensive lineman were eligible as the tackled ball carrier. Tackles that occurred within 5 yd of either sideline or the goal line were excluded. Both the ball carrier and defender could have distinctly different strategies in these locations, which would make them poor matches for tackles that occurred in more central field locations.

Only 1-on-1 tackles were included in this study. A tackle was determined to be 1-on-1 if a second defender did not assist in the tackle and was at least 2 yd away from the ball carrier. As a preliminary filter for this determination in the control cases, player position and velocity data from Next Gen Stats (NGS; NFL and Zebra Technologies) were used to exclude tackles when a second defensive player was within 2 yd of the ball carrier at the time of the tackle. The time of the tackle was provided in the NGS dataset. For both concussion and control cases, a video review was used to confirm that the tackle was 1-on-1. In some game situations, the ball carrier may go to the ground without any contact from the defender. For example, the ball carrier may go down intentionally so that the clock will continue to run, or he may go down unintentionally as a result of losing his balance. In either scenario, a simple touch of the ball carrier while on the ground by the defender is recorded as a tackle. Because these types of plays rarely result in an injury and are not consistent

*Address correspondence to Christopher P. Sherwood, MS, Biomechanics Consulting and Research, 1621 Quail Run, Charlottesville, VA 22911, USA (email: csherwood@biocorellc.com).

†Biomechanics Consulting and Research, Charlottesville, Virginia, USA.

‡School of Biomedical Sciences, University of Leeds, Leeds, UK.

§T.L. McMurry, LLC, Albany, CA, USA.

‖National Football League, New York, New York, USA.

¶Nanotechnology and Integrated Bioengineering Centre, School of Engineering, Ulster University, Belfast, UK.

Submitted June 24, 2024; accepted December 10, 2024.

One or more of the authors has declared the following potential conflict of interest or source of funding: The research presented in this article was made possible by a grant from the National Football League (NFL). The views expressed are solely those of the authors and do not represent those of the NFL. C.P.S. is an employee of Biomechanics Consulting and Research. T.L.M. and J.R.F. have received consulting fees from Biomechanics Consulting and Research. J.R.C. has an ownership interest in Biomechanics Consulting and Research, which has received consulting fees from the NFL, including those for the conduct of the study submitted for publication. A.S. is an employee of the NFL. AOSSM checks author disclosures against the Open Payments Database (OPD). AOSSM has not conducted an independent investigation on the OPD and disclaims any liability or responsibility relating thereto.

TABLE 1
Tackle Techniques and Characteristics Used for Qualitative Video Analysis[a]

	Player	Descriptor
Tackle type	T	Arm: tackler impedes/stops ball carrier with arm(s) Helmet: tackler uses helmet to strike ball carrier Shoulder: primary contact is with tackler's shoulder Smother: tackler uses chest and wraps both arms around ball carrier Torso: tackler impedes ball carrier using his torso only (no arms or shoulders)
Tackle direction	T	Behind: tackler contacts back of ball carrier (180° ± 30°) Side: tackler contacts side of ball carrier (90° ± 30°) Front: tackler contacts front of ball carrier (0° ± 30°)
Intended contact location on ball carrier	T	Arm, helmet, hip, lower leg, shoulder, torso, upper leg
Shortening steps (Y/N)	T	Tackler exhibited shorter and faster steps when approaching ball carrier
Active feet (Y/N)	T	Tackler's feet remained active throughout tackle
Head placed on correct side (Y/N)	T	Tackler attempted to position his helmet to side of ball carrier AND not into path of ball carrier
Foot width ≤ shoulder width (Y/N)	T	Tackler's foot width was less than or equal to shoulder width
Feet ≥ stride length apart (Y/N)	T	Tackler's feet were equal to or greater than stride length apart
Use of arms (Y/N)	T	Tackler used at least one arm to wrap up ball carrier
Head contact (Y/N)	T	Tackler's head trajectory changed noticeably because of direct contact with ball carrier
Leading with helmet (Y/N)	T/BC	Initial contact with top, front, or facemask (upper edge) of helmet
Initial contact location	T/BC	Arm, helmet, hip, lower leg, shoulder, torso, upper leg
Primary loading[b]	T/BC	Arm, helmet, hip, lower leg, shoulder, torso, upper leg
No. of feet on ground	T/BC	No. of feet on ground at time of tackle (0, 1, or 2)
Body position	T/BC	Falling/diving: falling or diving to ground Fully bent: greater than approximately 60° bend at waist Partially bent: approximately 30° to 60° bend at waist Upright: no bend at waist
Explosiveness (Y/N)	T/BC	Noticeable momentum transfer to other player
Use of helmet penalty (Y/N)	T/BC	Violation of current NFL use of helmet rule
Evasive maneuver (Y/N)	BC	Ball carrier made shuffle, spin, or side-step to attempt contact evasion
Stiff arm/fending (Y/N)	BC	Ball carrier utilized free arm to contact/repel tackler
Ball carrier aware (Y/N)	BC	Ball carrier was aware of impending tackle

[a]Coding was performed for the tackler (T), the ball carrier (BC), or both players (T/BC). NFL, National Football League; Y/N, yes/no.
[b]Primary loading: body part that sustained the most force.

with the type of dynamic tackles being investigated in this study, they were excluded during the video review process. There are approximately 21,000 tackles in an NFL season. Limiting these to 1-on-1 tackles away from the sideline and end zone reduces the number of tackles to approximately 7000 (~33%).

A concussive play was defined as a single play in which a diagnosed concussion was determined to have occurred as described previously in detail by Lessley et al[12] and Ellenbogen et al.[5] Concussive impact cases occurring outside the field of video coverage were excluded, as were cases in which there was uncertainty about which collision caused the injury. Concussions were selected from the 2015 to 2019 NFL seasons, including any preseason, regular season, or postseason games in which NGS player position data were available. NGS data were provided in 2 dimensions within the plane of the playing field: from goal line to goal line (x) and across the field (y). Concussions caused by ground contact were excluded, meaning that all concussions included in this study were caused by the tackler's helmet striking some part of the ball carrier.

Case-Control Matching

For the control dataset, potential tackles were selected from the 2017 to 2018 NFL seasons in games for which NGS player tracking data were available. Pro Football Focus data were used to identify the ball carrier and the tackler involved in successful tackles. Attempted/missed tackles were excluded. The objective was to select control cases that had similar characteristics to each concussion case. A video review of all potential control cases was not feasible; therefore, a method was developed to select a reduced sample of potential control cases using NGS data. Overall, 3 variables were chosen for this purpose: ball carrier trajectory, tackler trajectory, and closing speed between the players (calculated from the velocity [speed and vector] of both players). These were selected based on the importance of speed in the risk of concussions and to account for the many different tackling scenarios (head on, from the side, chasing from behind, etc). The time of the tackle for each play was provided in the NGS dataset. Trajectories were then calculated by finding the position vector defined by a player's NGS on-field location

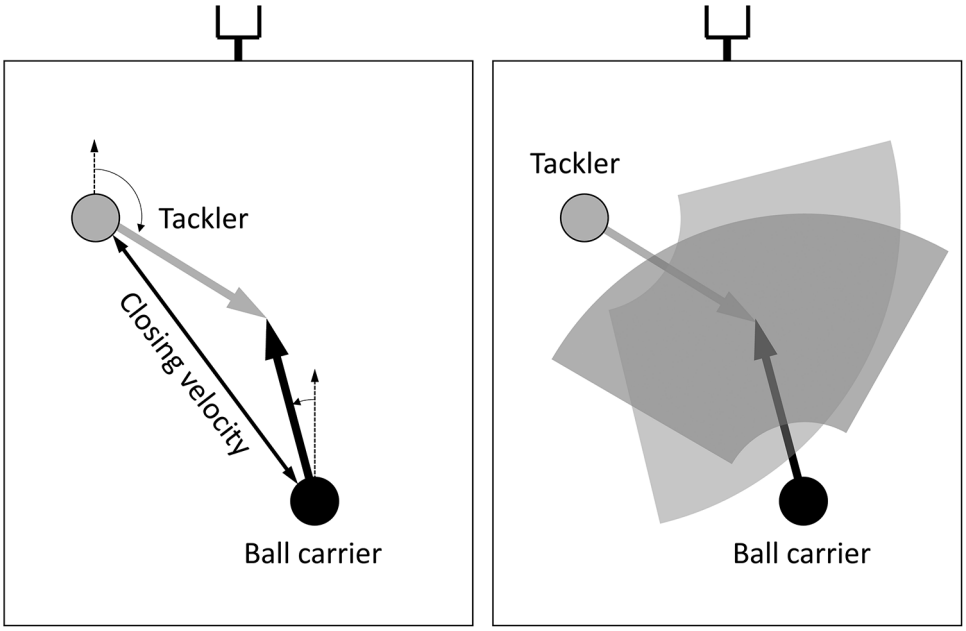

Figure 1. Next Gen Stats data were used to define the velocity vector of each player immediately before impact (left). Potential control cases were excluded if either player's trajectory deviated from the case player's trajectory by more than 45° or if the closing speed differed by more than 3 m/s (right).

from 0.3 seconds before the tackle until the time of the tackle. Both players' trajectories were measured relative to the downfield axis, with a clockwise rotation being defined as positive (Figure 1). Closing speed was calculated as the magnitude of the difference between the 2 players' velocity vectors at 0.3 seconds before the tackle. Because many tackles would be identical other than being mirrored on the opposite side of the field, trajectory angles for both players were negated for all plays in which the ball carrier had a leftward trajectory.

To ensure that controls had a minimum level of similarity, potential control cases were excluded as matches if they differed from the concussion case by more than 45° for either player's trajectory or if the closing speed differed by more than 3 m/s. For the remaining potential control cases, a similarity score was determined based on the root mean square error between the player trajectory and closing velocity in the case versus control. For each of the 3 variables (ball carrier trajectory, tackler trajectory, and closing speed), the difference between the concussion and control values was normalized by the difference between the maximum and minimum values of each variable. This effectively normalized the trajectory difference by up to 90° and the closing speed difference by up to 6 m/s. These 3 normalized differences were squared and summed, and then, the square root was taken. Potential control cases were reviewed in order of most to least similar (lowest to highest root mean square error). If the tackle involved the ball carrier falling or diving, the case was excluded. Additional control cases were reviewed until there were up to 3 matching controls. For some concussions, only 1 or 2 matching control cases could be identified.

Tackling Technique Coding

For all concussion and control cases, 20 tackle techniques and characteristics (Table 1) were coded based on previous video review studies of rugby tackles.[4,8,9,19,22,25] Some of the coding required binary (yes/no) determinations, whereas others required choosing from among several descriptors. The term "primary," used in many descriptors, was defined as the body part that sustained the most force. Tackle type was defined by the primary body part used by the tackler (Figure 2). The placement of the tackler's head was considered "correct" if it was away from the path of the ball carrier and "incorrect" if it was into the ball carrier's body or path (Figure 3). Of the 20 tackle techniques and characteristics, 10 were evaluated for the tackler only, 3 were evaluated for the ball carrier only, and 7 were evaluated for both. Therefore, 27 coding decisions were made for each case. For each case, a minimum of 2 camera views (minimum of 60 frames/s) were available: end zone and sideline views. Additional broadcast videos were viewed when additional footage was necessary. The NFL use of helmet rule was coded for all cases, even though it was not in effect for all seasons included in this study.

Statistical Analysis

The risk associated with tackling behavior was quantified using odds ratios (ORs) estimated from conditional logistic regression adjusted for closing velocity. Conditional logistic regression was chosen to account for the matched design of the dataset. Although closing velocity was included in the case-control matching criteria, the match was not perfect

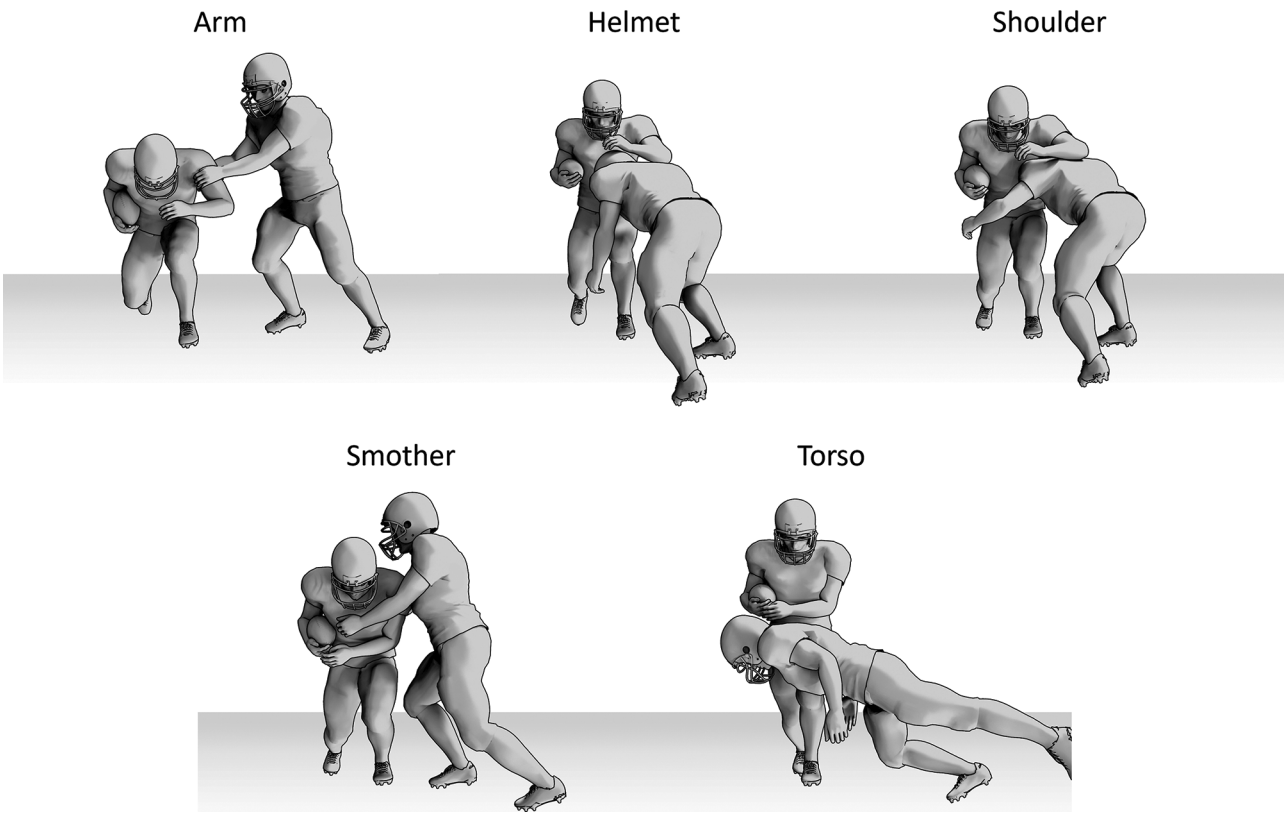

Figure 2. Tackle type defined by the primary body part used by the tackler.

(within 3 m/s). Because the concussion risk was thought to be strongly related to closing velocity, we decided to explicitly include closing velocity in the conditional logistic regression models to correct for any remaining systematic differences between cases and controls. Differences in the concussion risk by tackling behavior were assessed using the likelihood ratio test. A separate model for each tackling behavior was considered, and then, effects were compared across models using the FDR. For each tackle technique or characteristic, an unadjusted P value and an FDR-adjusted P value[1] were calculated to assess whether the different descriptors could differentiate between concussion and control cases. An FDR threshold of 0.10 was used to control for multiple comparisons. For each predictor, the OR and 95% confidence interval (CI) were calculated relative to a reference predictor. Reference predictors were generally selected based on them having the most common behavior.

RESULTS

A total of 51 concussion cases and 96 control cases were selected that met the inclusion criteria for the study. Overall, 18 of the concussion cases had 1 matched control case, 21 concussion cases had 2 matched controls, and 12 concussion cases had 3 matched controls. The distributions of play types and player positions in the selected concussion cases were generally similar to those in the selected control cases (Table 2) and to concussions in the NFL overall.[12] Coding was completed as described in the Methods section.

Conditional logistic regression modeling was performed for 23 of the 27 coded tackling techniques and characteristics (Table 3). In every model, the effect of closing speed was significant, with each 1-m/s increase in closing speed corresponding to roughly an order of magnitude increase in the concussion risk. In 4 models, no ORs could be calculated for the tackling characteristic because of the inadequate sample size. There were very few cases in which a use of helmet penalty was called; thus, the effect of that penalty could not be evaluated in 2 models. Likewise, the ball carrier anticipated the impending tackle and made contact with the tackler's head in every case; therefore, these factors could not be evaluated in another 2 models.

Of the remaining 23 models, 12 of the tackling techniques and characteristics had predictors that were significantly associated with a tackler's concussion. The tackler's risk of concussions was significantly affected by tackle type, with helmet tackles posing the highest risk; shoulder tackles posing a lower risk; and arm, torso, and smother tackles posing much lower risks. The lowest risk tackles occurred when the intended target was the ball carrier's torso. Tackle direction was an important factor, with front tackles posing more than twice the risk of a tackler's concussion than side tackles. There were no concussions in tackles from behind. When the tackler led with his helmet

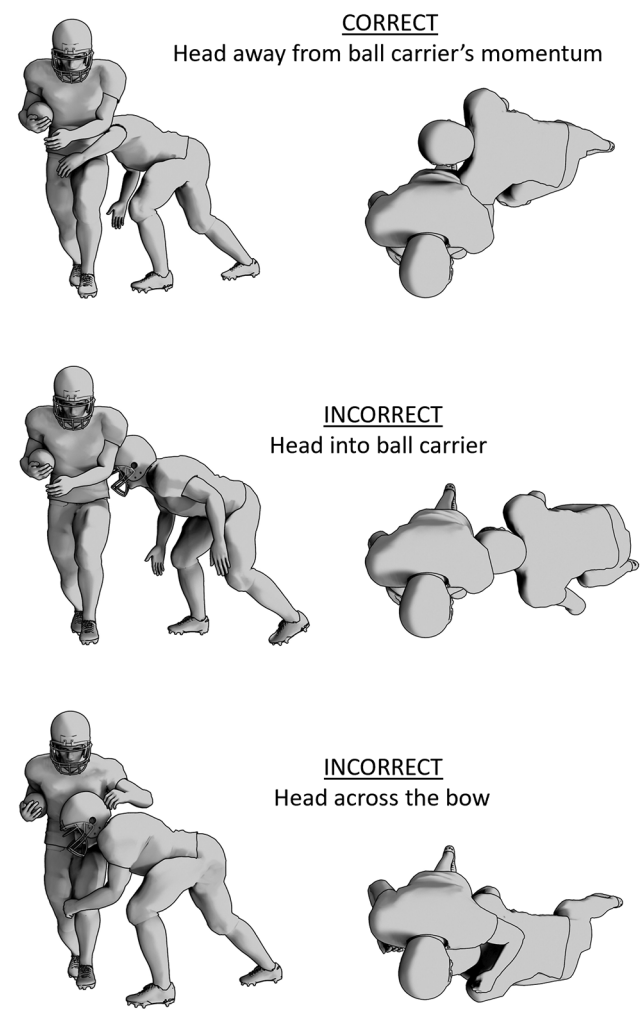

Figure 3. ''Correct'' and ''incorrect'' placement of the tackler's head in side tackles.

or when the initial or primary contact point was to the tackler's helmet, his risk of concussions was roughly 5 times higher than alternative scenarios. Interestingly, the tackler's risk of concussions was also significantly higher when the helmet of the ball carrier was the intended (initial or primary) contact point and nonsignificantly higher when the ball carrier led with his helmet. When the tackler used incorrect helmet placement (placing his helmet into the ball carrier or into the path of the ball carrier), his risk of concussions increased nearly 4-fold. Other significant variables included tackler foot width less than shoulder width, ball carrier explosiveness, and ball carrier making an evasive maneuver.

Several of the tackling techniques and characteristics examined did not have a statistically significant effect on the tackler's concussion risk. Logistic regression models suggested a higher risk of concussions when the tackler or ball carrier was partially bent over, but these results were not statistically significant. Most of the variables relating to the position and activity of the arms and feet were not significant. The tackler's concussion risk was not statistically significantly associated with either player's use of arms, tackler shortening steps, tackler active feet, the number of feet either player had on the ground, tackler explosiveness (noticeable momentum transfer to the ball carrier), or tackler feet being greater than stride length apart. However, a significantly elevated risk of concussions was observed when the tackler's feet were less than shoulder width apart.

TABLE 2
Distribution of Play Type and Player Position[a]

	Concussion (n = 51)	Control (n = 96)
Play type		
Rush	19 (37)	32 (33)
Pass	32 (63)	64 (67)
Ball carrier position		
Quarterback	1 (2)	5 (5)
Tight end	8 (16)	12 (13)
Wide receiver	10 (20)	36 (38)
Running back	32 (63)	43 (45)
Tackler position		
Defensive lineman	1 (2)	6 (6)
Linebacker	9 (18)	21 (22)
Defensive back	41 (80)	69 (72)

[a]Data are expressed as No. (%).

DISCUSSION

This matched case-control study using a qualitative video review was meant to identify tackling techniques and characteristics that pose an elevated risk of concussions for the tackling player in professional football. Although the ball carrier is often thought of as being in the more vulnerable position, the incidence of concussions in NFL games is nearly twice as high in tacklers compared with ball carriers.[12] The present study identified several tackle characteristics and techniques that had a significant effect on the tackler's risk of concussions. Closing speed had the most consistent and dramatic effect on the concussion risk, a finding that has been documented in several previous studies.[3,6,18,23,25,26] Front tackles presented a higher risk of concussions than side or rear tackles likely because they were associated with higher closing speeds. Beyond that, the risk of concussions generally increased with the degree of helmet involvement in the tackle. Helmet tackles, leading with the helmet, and initial or primary contact with the helmet were all factors associated with a significantly increased risk of tackler concussions. Correspondingly, the risk of tackler concussions generally decreased as the distance between the contact point and the head increased. Shoulder contacts presented a lower risk than helmet contacts, while torso and extremity contacts posed the lowest risks. These findings are similar to those in rugby union that discourage tackling high on the ball carrier's body.[24,25] When the tackler placed his head

TABLE 3
Conditional Logistic Regression Model Predictors[a]

	OR (95% CI)	*P* Value
Tackle type		
Shoulder	Reference	
Arm	0.11 (0.02-0.82)	.0101 (unadjusted)
Helmet	2.40 (0.17-34.64)	**.0258 (FDR)**
Smother	0.43 (0.12-1.54)	
Torso	0.09 (0.01-0.99)	
Closing velocity	11.35 (2.68-48.17)	
Tackle direction		
Side	Reference	
Front	2.37 (0.80-7.05)	.005 (unadjusted)
Closing velocity	10.13 (3.02-33.97)	**.0164 (FDR)**
Intended contact on ball carrier		
Shoulder	Reference	
Helmet	1.10 (0.15-7.88)	.0377 (unadjusted)
Torso	0.18 (0.05-0.69)	**.0788 (FDR)**
Upper leg	0.40 (0.12-1.33)	
Closing velocity	15.29 (3.54-66.09)	
Shortening steps (Y/N)		
No	Reference	
Shortening steps	0.91 (0.36-2.32)	.8416 (unadjusted)
Closing velocity	9.84 (3.01-32.21)	.8416 (FDR)
Active feet (Y/N)		
No	Reference	
Active feet	0.71 (0.29-1.75)	.4545 (unadjusted)
Closing velocity	9.29 (2.86-30.20)	.5501 (FDR)
Tackler leading with helmet (Y/N)		
No	Reference	
Leading with helmet	5.18 (1.87-14.33)	.0004 (unadjusted)
Closing velocity	9.59 (2.67-34.37)	**.0032 (FDR)**
Tackler initial contact		
Helmet	Reference	
Shoulder	0.17 (0.04-0.78)	.0000 (unadjusted)
Closing velocity	6.62 (1.66-26.36)	**.0001 (FDR)**
Tackler primary loading		
Shoulder	Reference	
Helmet	4.61 (1.41-15.05)	.0000 (unadjusted)
Closing velocity	9.23 (2.06-41.28)	**.0000 (FDR)**
Tackler No. of feet on ground		
0	Reference	
1	5.82 (0.46-73.36)	.2519 (unadjusted)
2	6.20 (0.56-68.76)	.3219 (FDR)
Closing velocity	10.99 (3.17-38.07)	
Foot width $\leq$ shoulder width (Y/N)		
No	Reference	
Foot $\leq$ shoulder	0.24 (0.08-0.74)	.0082 (unadjusted)
Closing velocity	17.55 (4.06-75.94)	**.0235 (FDR)**
Feet $\geq$ stride length apart (Y/N)		
No	Reference	
Feet $\geq$ stride length	1.33 (0.53-3.38)	.5417 (unadjusted)
Closing velocity	10.75 (3.14-36.76)	.623 (FDR)
Tackler body position		
Partially bent	Reference	
Falling/diving	0.36 (0.08-1.69)	.2215 (unadjusted)
Fully bent	1.13 (0.40-3.20)	.2996 (FDR)
Upright	0.23 (0.02-3.25)	
Closing velocity	12.13 (3.19-46.07)	
Tackler explosiveness (Y/N)		
No	Reference	
Explosiveness	0.86 (0.37-1.96)	.715 (unadjusted)
Closing velocity	9.65 (2.95-31.50)	.7831 (FDR)

(continued)

TABLE 3
(continued)

	OR (95% CI)	*P* Value
Head placed on correct side (Y/N)		
Yes	Reference	
Head across bow	4.17 (1.58-11.01)	.0015 (unadjusted)
Closing velocity	8.64 (2.47-30.31)	**.0088 (FDR)**
Ball carrier evasive maneuver (Y/N)		
No	Reference	
Evasive maneuver	0.39 (0.14-1.05)	.0512 (unadjusted)
Closing velocity	10.02 (2.95-34.05)	**.0981 (FDR)**
Ball carrier leading with helmet (Y/N)		
No	Reference	
Leading with helmet	3.45 (0.86-13.81)	.0722 (unadjusted)
Closing velocity	9.90 (3.06-32.01)	.1242 (FDR)
Ball carrier initial contact		
Upper leg	Reference	
Arm	0.98 (0.16-5.92)	.003 (unadjusted)
Helmet	5.11 (0.90-29.08)	**.0114 (FDR)**
Hip	1.03 (0.13-8.00)	
Shoulder	1.31 (0.33-5.11)	
Closing velocity	9.19 (2.49-33.89)	
Ball carrier primary loading		
Torso	Reference	
Helmet	33.86 (2.74-418.71)	.0024 (unadjusted)
Hip	5.23 (0.27-100.65)	**.011 (FDR)**
Lower leg	2.15 (0.06-82.21)	
Shoulder	17.77 (1.88-167.70)	
Upper leg	4.23 (0.43-41.25)	
Closing velocity	10.91 (2.83-42.05)	
Ball carrier No. of feet on ground		
0	Reference	
1	0.68 (0.15-3.17)	.7973 (unadjusted)
2	0.54 (0.09-3.23)	.8335 (FDR)
Closing velocity	10.46 (3.12-35.06)	
Ball carrier body position		
Upright	Reference	
Fully bent	1.34 (0.30-5.93)	.0756 (unadjusted)
Partially bent	2.83 (1.06-7.52)	.1242 (FDR)
Closing velocity	10.12 (2.90-35.34)	
Ball carrier stiff arm/fending (Y/N)		
No	Reference	
Stiff arm/fending	0.18 (0.02-1.54)	.0812 (unadjusted)
Closing velocity	11.49 (3.31-39.93)	.1246 (FDR)
Ball carrier explosiveness (Y/N)		
No	Reference	
Explosiveness	3.99 (1.08-14.74)	.0303 (unadjusted)
Closing velocity	9.71 (2.79-33.75)	**.0698 (FDR)**
Tackler use of arms (Y/N)		
No	Reference	
Use of arms	0.53 (0.20-1.40)	.1962 (unadjusted)
Closing velocity	10.04 (3.06-32.92)	.2821 (FDR)

[a]Results for predictors with undefined ORs were omitted. Statistically significant results (FDR-adjusted $P < .10$) are bolded. FDR, false discovery rate; N, no; OR, odds ratio; Y, yes.

on the correct side of the ball carrier, his risk of concussions was cut by 74%.

Some of the statistically significant associations uncovered by conditional logistic regression modeling could have resulted from correlations with other variables. For example, it is not clear why tackles with a narrow foot width by the tackler were associated with an increased risk of concussions. However, it turned out that the tackler's feet were most often shoulder width apart or less in shoulder tackles, helmet tackles, and front tackles, all of which were associated with an elevated risk of tackler concussions on their own. In addition, although it is obvious

why the tackler's risk of concussions was significantly higher with contact to his own helmet, it is less obvious why the tackler's risk of concussions was elevated when the ball carrier's helmet was contacted. It may be because of the stiffness of the ball carrier's helmet, but other factors such as effective mass also play a role. We observed that the ball carrier's helmet was struck most often in situations in which the risk of tackler concussions was elevated, that is, in front tackles, when the ball carrier led with his helmet and when the tackler placed his helmet on the incorrect side of the ball carrier.

A qualitative video review has been used extensively to examine tackling techniques in rugby union, which utilizes tackling techniques similar to those in football.[2,9,20,21,25] These studies have identified several patterns of tackler behavior associated with a reduced rate of head injuries for the tackler: shortening steps before contact[22]; bending at the waist[25]; tackler's use of arms[22]; and tackler placing his head on the "correct" side of the ball carrier, away from the ball carrier's forward momentum.[21,22] Unlike similar studies in rugby union,[22,25] the present study of NFL tackles did not find significant associations between tackler concussions and shortening steps before contact, bending at the waist, or the tackler's use of his arms. However, an important finding from rugby union[21,22] replicated in the present study is that the tackler significantly reduced his risk of concussions by placing his head on the correct side of the ball carrier, out of the path of the ball carrier's forward momentum.

There were a number of limitations to this study. The study was limited to 1-on-1 tackles not occurring near the sideline or goal line. Although efforts were made to match concussion cases with controls based on player trajectory and closing speed, no match was perfect because of the wide variety of tackles occurring on the field. A number of the tackle techniques and characteristics that were evaluated were related to each other and not independent (eg, "helmet tackle" and "leading with the helmet"). Similar results are therefore to be expected between certain variables. Other potentially important variables were not coded, such as game clock time and down number, because they may have played a role in the ball carrier's decision process (eg, staying inbounds). There were also several tackling behaviors that were infrequent in either the concussion or control group, which can result in degraded performance of the underlying statistical models.

In spite of these limitations, this study produced convincing evidence that the less a tackler involves his helmet in a tackle, the lower his risk of sustaining a concussion. In particular, this study supports teaching players that they should aim for the torso of the ball carrier and place their heads out of the path of the ball carrier when tackling to reduce their risk of concussions.

ACKNOWLEDGMENT

The authors thank the NFL Players Association, the NFL Player Health and Safety Department, NFL Films for the provision of video footage, Rachel E. Michaels for her assistance in manuscript formatting, and Pro Football Focus for providing NFL game variables.

ORCID iDs

Christopher P. Sherwood https://orcid.org/0009-0001-8837-9232
Fintan Grogan https://orcid.org/0009-0002-0973-8346
Timothy L. McMurry https://orcid.org/0000-0001-5912-5092
James R. Funk https://orcid.org/0000-0002-5329-8352
Allen Sills https://orcid.org/0000-0003-1145-0480
Gregory Tierney https://orcid.org/0000-0002-4666-4473

REFERENCES

1. Altman DG. *Practical Statistics for Medical Research*. CRC Press; 1990.
2. Burger N, Lambert MI, Viljoen W, Brown JC, Readhead C, Hendricks S. Tackle technique and tackle-related injuries in high-level South African Rugby Union under-18 players: real-match video analysis. *Br J Sports Med*. 2016;50(15):932-938.
3. Cross MJ, Tucker R, Raftery M, et al. Tackling concussion in professional rugby union: a case-control study of tackle-based risk factors and recommendations for primary prevention. *Br J Sports Med*. 2019;53(16):1021-1025.
4. Davidow D, Quarrie K, Viljoen W, et al. Tackle technique of rugby union players during head impact tackles compared to injury free tackles. *J Sci Med Sport*. 2018;21(10):1025-1031.
5. Ellenbogen RG, Batjer H, Cardenas J, et al. National Football League Head, Neck and Spine Committee's concussion diagnosis and management protocol: 2017-18 season. *Br J Sports Med*. 2018;52:894-902.
6. Fuller CW, Ashton T, Brooks JH, Cancea RJ, Hall J, Kemp SP. Injury risks associated with tackling in rugby union. *Br J Sports Med*. 2010;44(3):159-167.
7. Fuller CW, Junge A, Dvorak J. A six year prospective study of the incidence and causes of head and neck injuries in international football. *Br J Sports Med*. 2005;39(Suppl 1):i3-i9.
8. Hendricks S, O'Connor S, Lambert M, et al. Contact technique and concussions in the South African under-18 Coca-Cola Craven Week rugby tournament. *Eur J Sport Sci*. 2015;15(6):557-564.
9. Hendricks S, Till K, Hollander S, et al. Consensus on a video analysis framework of descriptors and definitions by the Rugby Union Video Analysis Consensus Group. *Br J Sports Med*. 2020;54(10):566-572.
10. Hutchinson MG, Comper P, Meeuwisse WH, Echemendia RJ. A systematic video analysis of National Hockey League (NHL) concussions, part II: how concussions occur in the NHL. *Br J Sports Med*. 2015;49(8):552-555.
11. Jacobi J, Wasserman EB, Mack CD, et al. The National Football League concussion protocol: a review. *HSS J*. 2023;19(3):269-276.
12. Lessley D, Kent R, Funk J, et al. Video analysis of reported concussion events in the National Football League during the 2015-2016 and 2016-2017 seasons. *Am J Sports Med*. 2018;46(14):3502-3510.
13. Maerlender A, Masterson CJ, Norris R, Hinthorne A. Validating tackle mechanics in American football: improving safety and performance. *Ann Biomed Eng*. 2020;48(11):2691-2700.
14. NFL Football Operations. 2020 NFL-NFLPA CBA (collective bargaining agreement): need to know. Available at: https://operations.nfl.com/inside-football-ops/players-legends/2020-nfl-nflpa-cba-need-to-know/. Accessed May 1, 2024.
15. NFL Football Operations. 2023 NFL Rulebook: 2023 rule changes. Available at: https://operations.nfl.com/the-rules/nfl-rulebook/#2023-rule-changes. Accessed May 1, 2024.
16. NFL Football Operations. 2023 NFL Rulebook: 2023 official playing rules of the National Football League. Rule 3: definitions. Section

35: tackling. Available at: https://operations.nfl.com/the-rules/nfl-rulebook/#section-35-tackling. Accessed May 1, 2024.
17. Pellman E, Viano D, Tucker A, Casson I. Concussion in professional football: location and direction of helmet impacts. Part 2. *Neurosurgery*. 2003;53(6):1328-1341.
18. Quarrie KL, Hopkins WG. Tackle injuries in professional rugby union. *Am J Sports Med*. 2008;36(9):1705-1716.
19. Stockwell DW, Blalock R, Podell K, Marco RAW. At-risk tackling techniques in American football. *Orthop J Sports Med*. 2020;8(2):2325967120902714.
20. Stokes KA, Locke D, Roberts S, et al. Does reducing the height of the tackle through law change in elite men's rugby union (the Championship, England) reduce the incidence of concussion? A controlled study in 126 games. *Br J Sports Med*. 2021;55(4):220-225.
21. Suzuki K, Nagai S, Iwai K, Furukawa T, Takemura M. How does the situation before a tackle influence a tackler's head placement in rugby union? Application of the decision tree analysis. *BMJ Open Sport Exerc Med*. 2021;7(1):e000949.
22. Tierney GJ, Denvir K, Farrell G, Simms CK. The effect of tackler technique on head injury assessment risk in elite rugby union. *Med Sci Sports Exerc*. 2018;50(3):603-608.
23. Tierney GJ, Lawler J, Denvir K, McQuilkin K, Simms CK. Risks associated with significant head impact events in elite rugby union. *Brain Inj*. 2016;30(11):1350-1361.
24. Tierney GJ, Simms CK. Can tackle height influence head injury assessment risk in elite rugby union? *J Sci Med Sport*. 2018;21(12):1210-1214.
25. Tucker R, Raftery M, Kemp S, et al. Risk factors for head injury events in professional rugby union: a video analysis of 464 head injury events to inform proposed injury prevention strategies. *Br J Sports Med*. 2017;51(15):1152-1157.
26. Viano DC, Pellman EJ. Concussion in professional football: biomechanics of the striking player. Part 8. *Neurosurgery*. 2005;56(2):266-280.

For reprints and permission queries, please visit Sage's Web site at http://www.sagepub.com/journals-permissions